



Course Description and Outcome Form
Department of Computer Science and Engineering
School of Data and Sciences
BRAC University

A. Course General Information:

Course Code:	CSE 340
Course Title:	Computer Architecture
Credit Hours (Theory + Lab):	3+0
Contact Hours (Theory + Lab):	3+0
Category:	Program Core
Type:	Required, Engineering, Lecture
Prerequisites:	CSE260 (Hard Prerequisite)
Co-requisites:	None

B. Course Catalog Description (Content):

A systematic study of the various elements in computer design, including circuit design, storage mechanisms, addressing schemes, and various approaches to parallelism and distributed logic. Information representation and transfer; instruction and data access methods; the control unit; hardware and microprogrammed; memory organization; RISC and CISC machines.

C. Course Objective:

The objectives of this course are:

- a) Introduce different processor technologies, performance metrics, representation of numbers, and arithmetic operations.
- b) Introduce RISC-V architecture, demonstrate its instruction formats and translation of simple C/Java code snippets to RISC-V assembly language.
- c) Teach how to design processor datapaths, recognize pipelining hazards, and use different techniques to overcome them.
- d) Introduce and explain memory hierarchy and performance analysis.
- e) Introduce parallel architecture and parallel programming.

D. Course Outcomes (COs):

Upon completing this course, students will be able to:

Sl.	CO Description	Weightage (%)
CO1	Explain the processor technologies and performance metrics alongside the representation of numbers and arithmetic operations	20
CO2	Demonstrate various instruction formats, their encoding, translation from C/Java code to RISC-V instruction	25
CO3	Visualize the datapath of different instructions and recognize various pipelining hazards and hazard overcoming techniques	20
CO4	Outline memory hierarchy and performance analysis and review various parallel architecture and its programming	10
CO5	Evaluate the sustainability and impact of modern fast-changing computer architecture and technology in the society and environment	5

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Explain the processor technologies, performance metrics alongside the representation of numbers and arithmetic operations	a	Cognitive	Lectures, slides, notes	Quiz, Assignment, Midterm, Final
CO2	Demonstrate various instruction formats, their encoding, translation from C/Java code to RISC-V instruction	a	Cognitive	Lectures, slides, notes	Quiz, Assignment, Midterm, Final
CO3	Visualize the datapath of different instructions and recognize various pipelining hazards and hazard overcoming techniques	b	Cognitive	Lectures, slides, notes	Quiz, Assignment, Midterm, Final
CO4	Outline memory hierarchy and performance analysis and review various parallel architecture and its programming	a	Cognitive	Lectures, slides, notes	Quiz, Final
CO5	Evaluate the sustainability and impact of modern fast-changing computer architecture and technology in the society and environment	g	Cognitive		Report Submission

F. Course Materials:**i. Text and Reference Books:**

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Computer Organization and Design RISC-V edition	D. A. Patterson, J. L. Hennessy		2 nd Edition	Morgan Kaufmann	9780128203316

ii. Other materials (if any)

Lecture Notes, Slides, Practice Sheets

G. Lesson Plan:

No	Topic	Week/Lecture #	Related CO (if any)
1	Various Computer Architecture Models, Measuring Performance CPI, Performance Equation, RISC and CISC Architecture	1,2	CO1
2	Introduction of the Computer Architecture, Introduction to RISC-V Assembly Language, Instruction sets, Conversion from High-level code to RISC-V equivalent form, how function call works, etc.	2, 3, 4, 5	CO2
3	Arithmetic for Computers: Addition, Subtraction, Multiplication, Division, IEEE-754 floating-point conversion, floating-point operations in RISC-V architecture	6, 7	CO1
Midterm			
4	RISC-V non-pipelined datapath/control path, how single-cycle datapath works for various instruction, advantages, disadvantages of single-cycle datapath	8	CO3
4	RISC-V pipelined datapath, how it works, advantages and disadvantages, designing pipelined datapath, Hazard and hazard overcoming techniques	9	CO3
5	Memory Hierarchy, Memory Technology, Cache Introduction and Improving Cache Performance, Virtual memory, Cache Miss	10, 11	CO4
6	Introduction of Parallel Processors, SISD, MIMD, SIMD, SPMD, and Vector, Multicore, multi-threading, GPU	12, 13	CO4
Final			

H. Assessment Tools:

Assessment Tools	Weightage (%)
1. Participation in class (Attendance)	5
2. Research Report	5
3. Quizzes/Class Tests	15
4. Assignments	10
5. Mid Term Examination	25
6. Final Examination	40
Total	100%

I. CO Assessment Plan:

Assessment Tools	Course Outcomes				
	CO1	CO2	CO3	CO4	CO5
1. Quizzes/Class Tests	√	√	√	√	
2. Mid-Term Examination	√	√			
3. Assignments	√	√	√		
4. Research Report					√
5. Final Exam	√	√	√	√	

J. CO Attainment Policy:

As per Department of CSE Course Outcome Attainment Policy

K. Grading policy:

As per BRAC University grading policy

L. Course Coordinator:

Partha Bhoumik (PBK)